

Autonomous QA Agent

AI Quality Engineering Case Study

Role	AI Quality Engineer Solution Architect Lead Developer
Duration	Independent AI Engineering Project
Domain	AI Quality Engineering Agentic AI Test Automation

Business Problem

Enterprise UI automation becomes expensive to maintain due to brittle locators, DOM changes, flaky executions, and manual failure analysis. These issues increase maintenance effort and slow release cycles.

Solution

Designed and built an Agentic AI-powered Autonomous QA Agent that generates Playwright tests from requirements, executes them, diagnoses failures, inspects the live DOM using Playwright MCP, performs AI-assisted locator healing, retries execution, and stores validated selectors for future reuse.

High-Level Workflow

Requirement → LLM → Playwright Test Generation → Execute → Failure Analysis → Playwright MCP → DOM Analysis → LLM Locator Healing → Retry → Report

Business Value

- Reduced manual test maintenance
- Faster failure diagnosis
- Improved regression stability
- Reusable healed locators
- Better CI/CD reliability
- Higher engineering productivity

Technology Stack

Agentic AI • LangChain • Playwright MCP • Groq LLM • DeepEval • TypeScript • Node.js • Playwright • GitHub

Project Highlights

- Autonomous test generation
- AI-assisted self-healing
- Live DOM inspection
- Intelligent retry mechanism
- Persistent selector knowledge base

Links

GitHub: <https://github.com/shaik-rahaman/autonomous-qa-agent>

Demo: <http://autonomousqa.ddns.net/>

Summary

This project demonstrates how AI Quality Engineering can reduce repetitive QA work by combining LLM reasoning, browser tooling, and autonomous execution into a practical enterprise-style automation workflow.